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Exercise 10
IHW#6: Your firm - Ratio computation and analysis
Aka: Your firm's ratios

The purpose of this HW is to enhance your ability to (1) compute and interpret some commonly used financial ratios, (2) understand the implications of the ratios, and (3) develop skills in using ratios to make inference about potential business/accounting issues and in the process gain a deeper understanding of firms and industries.

Two Readings

1. "Reading for Ratio HW"
2. Chapter 10 "Ratio analysis & its role in interpreting financial statements" *Financial Accounting: First Take*.

Materials: (1) the annual report and financial statements from your chosen firm and (2) other resources you find useful.

The ratios listed below and their formulas are shown in the table attached. You will be asked to calculate the ratios.

Section 2: Ratios using data from income statements only

Ratio 1: Gross profit percentage = Gross margin / Net sales

Ratio 2: Operating income percentage = Operating income / Net sales

Ratio 3: Income tax rate = Provision for income taxes / Income before income taxes and minority interests

Ratio 4: Net operating profit after taxes (NOPAT) percentage = NOPAT / Net sales

Ratio 5: Net income percentage = Net income / Net sales

Ratio 6: Interest coverage ratio = Income before interest and taxes / Interest expense

Section 3: Ratios using data from balance sheets only

Ratio 7: Current ratio = Current assets / Current liabilities

Ratio 8: Financial leverage ratio = Total assets / Stockholders' equity

Section 4: Ratios using data from both income statements and balance sheets

Ratio 9: Total asset turnover (TAT) = Net sales / Total assets

Ratio 10: Return on assets (ROA) = NOPAT / Total assets

Ratio 11: Return on equity (ROE) = Net income / Stockholders' equity

Ratio 12: Days' inventory held = $365 / [\text{Cost of goods sold} / \text{Inventory}]$

Ratio 13: Days' A/R outstanding = $365 / [\text{Net sales} / \text{Accounts receivable}]$

Ratio 14: Days' A/P outstanding = $365 / [\text{Purchases} / \text{Accounts payable}]$

Ratio 15: Days' in financing = Days inventory held + days A/R outstanding – days A/P outstanding

Requirements:

1. Record the ratio formula in cell (1).
2. Before beginning, think about the intuition of each ratio and formulate an expected value for the ratios for THE FIRM. Your expected ratio value could be based on what you know about THE FIRM and its industry (or an educated guess). Record your expectations in cell (2) of the table below prior to calculating the ratios.
3. Calculate the values for each ratio (showing all formula and calculation) for the two most recent financial statement dates and record in cells (3) and (4) in the table below.
4. Compute the changes over the two periods and record in cell (5) in the table below.

- Briefly explain possible reasons for the changes across the two periods and record in cell (6) in the table below.
- Compare the actual ratio calculations with your expected values and record in cell (7) in the table below.

The goal of this problem is not to focus on the minutia of the calculations (although you should be familiar with the calculations), but to focus on the intuition and the inference you can draw from the calculations and the changes in the ratio values.

Your firm - Ratio computation and analysis

Ratio 1: Gross Profit Percentage = Gross Margin / Net Sales				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
GP% = (Revenue – cost of revenue)/Revenue	~66%	GP% = (60922-16621)/60922 = 72.72%	GP% = (26974-11618)/26974 = 56.93%	15.79%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio is primarily a measure of a firm's efficiency and their value in the industry. For NVIDIA even a 56% margin is very positive but the 72% value from 2024 shows how much power they have as sellers with their highly differentiated product.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. 72% is still substantially higher than my estimate for NVIDIA's success knowing how this year has gone. Their 2023 number is smaller than I would have anticipated given that their boom began in 2022 in my memory.				

Ratio 2: Operating Income Percentage = Operating Income / Net Sales				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Operating Income Percentage = Operating Income / Net Sales	~ 60%	Op Inc % = 32972/60922 = 54.12%	Op Inc % = 4224/26974 = 15.66 %	38.46%
(6) Discuss how this ratio provides insights into the firm and the firm's industry The ratio here shows how much of direct sales revenue is retained after removing the cost of good sold and income from non operating financials.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. These numbers are lower than I expected. It shows NVIDIA makes more money in investment that I had anticipated and in financing.				

Ratio 3: Income Tax Rate = Provision for Income Taxes / Income before Income Taxes and Minority Interests				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Income Tax Rate = Provision for Income Taxes / Income before Income Taxes and Minority Interests	~10%	Income tax rate = 4058/ 33918 = 12%	Income tax rate = -187/ 4181 = -4.47%	16.47%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio shows us the tax rate a company faces and how much tax pressure is put on this industry				
(7) Compare the actual ratio calculations with your expected values from part 2 above. The tax ratios is very different YoY due to the scale of the revenue change and how much of it came from sales.				

Ratio 4: Net Operating Profit after Taxes (NOPAT) Percentage = NOPAT / Net Sales				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
(NOPAT) Percentage = NOPAT / Net Sales	~45%	NOPAT% = 29760/60922 = 48.85%	NOPAT% = 4368/26974 = 16.19%	32.66%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio shows us more directly how much revenue comes from operating with a tax normalization that makes this value more comparable to companies with different tax structures.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. The NOPAT% calculated is very close to my expectation for NVIDIA after considering the OpInc%				

Ratio 5: Net Income Percentage = Net Income / Net Sales				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Net Income Percentage = Net Income / Net Sales	~ 45%	NI% = 29760 / 60922 = 48.85%	NI% = 4368 / 26974 = 16.19%	32.66%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio reflects to us how much of a companies revenue is kept in net income after interest, and taxes have been taken out. It is a better picture within the industry of how a company is competing with other companies with similar tax structures.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. Given what we already know about NOPAT this percentage is nearly right on my expected values for 2024.				

Ratio 6: Interest Coverage Ratio = Income before Interest and Taxes / Interest Expense				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Interest Coverage Ratio = Income before Interest and Taxes / Interest Expense	ICR = ~ 500%	ICR = 32972 / 257 = 12829.57%	ICR = 4224 / 262 = 1612.21%	11217.36%
(6) Discuss how this ratio provides insights into the firm and the firm's industry The interest coverage ratio tells us weather a companies income is enough to safely cover the expenses of any financing they have taken. In NVIDIA's case, this ratio shows that NVIDIA is more than capable of paying the interest on any loans and are operating with a tremendous margin to their interest expense.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. The ICR unbelievable high for 2024. Knowing that NVIDIA does utilize financing, I would have expected a higher interest expense.				

Ratio 7: Current Ratio = Current Assets / Current Liabilities				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Current Ratio = Current Assets / Current Liabilities	Current ratio ~ 250%	Current ratio = 44345 / 22750 = 194.92%	Current Ratio = 23073 / 19081 = 120.92%	74.00%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio give us a picture of how healthy a firm's financials are and shows their total liquid value vs. what they owe. In NVIDIA's case, they are showing an extremely positive ratio even before their 2024 boom. (7) Compare the actual ratio calculations with your expected values from part 2 above. The ratio value is slightly lower than the massive number I guessed at but still fairly close. NVIDIA's assets greatly outweigh their liabilities.				

Ratio 8: Financial Leverage Ratio = Total Assets / Stockholders' Equity				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Financial Leverage Ratio = Total Assets / Stockholders' Equity	~ 200%	FLR = 65728 / 42978 = 157.59%	FLR = 41182 / 22101 = 186.34%	(28.75%)
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio shows the amount of assets retained by the firm vs the amount the shareholders own. For NVIDIA this ratio is still firmly in favor of the Firm. (7) Compare the actual ratio calculations with your expected values from part 2 above. This ratio turned out somewhat lower than expected but it just highlights how healthy NVIDIA's assets are even with the boom in income from investment in 2024. The number went down from '23 into '24 due to the investment surge.				

Ratio 9: Total Asset Turnover (TAT) = Net Sales / Total Assets				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Total Asset Turnover (TAT) = Net Sales / Total Assets	~ 90 %	= 60922 / 67728 = 89.95%	= 26974 / 41182 = 65.50%	24.45%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio will tell us how easily a company could replace its total assets via its sales.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. NVIDIA's 2024 TAT is incredibly strong and explains how NVIDIA can post greater than 100% growth. They brought in sales revenue greater than 89% of what they currently own. The calculated value is very close to my estimation.				

Ratio 10: Return on Assets (ROA) = NOPAT / Total Assets				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Return on Assets (ROA) = NOPAT / Total Assets	40%	= 29760 / 67728 = 43.94%	= 4368 / 41182 = 10.61%	33.33%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio shows us that NVIDIA is making ~ 40% on the value of their assets yoy from operating alone. This reflects just how lucrative their business is as it is returning cash at a high rate from what it owns.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. The ROA calculated matches very close to my expectations which were guided by the knowledge of gross margin in some ways. I assumed NVIDIA would not have more in assets than they roughly make on sales.				

Ratio 11: Return on Equity (ROE) = Net Income / Stockholders' Equity				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Return on Equity (ROE) = Net Income / Stockholders' Equity	50 %	ROE = 29760 / 42978 = 69.24%	ROE = 4368 / 22101 = 19.76%	= 49.48%
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio shows us how much cash a firm generates based on the amount of equity sold to shareholders. For NVIDIA, the company is turning an excellent profit from the invested funds.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. The ROE is excellent. Even with a massive increase in SE the firm is generating tons of income by comparison.				

Ratio 12: Days' Inventory Held = 365 / [Cost of Goods Sold / Inventory]				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Days' Inventory Held = 365 / [Cost of Goods Sold / Inventory]	30 Days?	= 365 / (16621/5282) = 116 days	= 365 / (11618 / 2159) = 68 days	48 days
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio gives us an understanding of how a firm operates and may be used to estimate what a sales surge could look like for them based on how many days of inventory they have on hand.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. I tried to estimate this based off my knowledge of the dishwasher industry from my past. NVIDIA keeps higher inventory than I expected though I imagine that may be a factor of their product average size being smaller and easier to store?				

Ratio 13: Days' A/R Outstanding = $365 / [\text{Net sales} / \text{Accounts Receivable}]$				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Days' A/R Outstanding = $365 / [\text{Net sales} / \text{Accounts Receivable}]$	90 days?	Days AR Outstanding = $365 / (60922 / 9999) = 59.91$ Days	Days AR Outstanding = $365 / (26974 / 3827) = 51.79$ Days	8.12 Days
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio tells us how fast a firm is able to correct its accounts receivable. Since NVIDIA's is a normal length and hasn't changed much. This ratio doesn't tell us much about NVIDIA in the last two years.				
(7) Compare the actual ratio calculations with your expected values from part 2 above. NVIDIA holds 60 day payment terms as far as I can tell. While this is shorter than my estimate it is not really telling us anything new.				

Ratio 14: Days' A/P Outstanding = $365 / [\text{Purchases} / \text{Accounts Payable}]$				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Days' A/P Outstanding = $365 / [(\text{COR (Ending inventory - beginning Inventory)}) / \text{Accounts Payable}]$	90 days?	$= 365 / (51907383 / 2699) = .02$ days	$= 365 / (3868794 / 1193) = .11$ days	-.09 days
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio is a firm that shows how quickly a firm handles its accounts payable. It seems that NVIDIA pays their AP's typically in less than a day?				
(7) Compare the actual ratio calculations with your expected values from part 2 above. This number really surprises me? I don't think its possible to have accounts payable covered within a day				

Ratio 15: Days' in Financing = Days Inventory Held + Days A/R Outstanding – Days A/P Outstanding				
(1) Ratio formula	(2) Expected value for the ratio	(3) Calculations and values for the most recent year	(4) Calculations and values for the prior year	(5) Changes over the two periods
Days' in Financing = Days Inventory Held + Days A/R Outstanding – Days A/P Outstanding	100 days	= 116 + 60 – 0 = 176	= 68 + 52 – 0 = 120 days	56 days
(6) Discuss how this ratio provides insights into the firm and the firm's industry This ratio tells us how fast a firm can turn inventory into cash				
(7) Compare the actual ratio calculations with your expected values from part 2 above. This ratio shows us that NVIDIA has a fairly high throughput from inventory to cash. With days in financing < 180, NVIDIA can flip their entire inventory into cash more than twice a year.				